

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read: at the patch order of a simulated cortical sheet it constrains at most 10 sites, so on a larger population the positive link from coherence to content rests entirely on a bridge assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps

and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows consistency, not cortical realization. The conditions yield a disconfirmable comparison: the agreement of decoders fitted independently on two regions, in excess of what their separate accuracies imply, should be larger for perceived than for unperceived stimuli at matched phase coherence and single-region accuracy, with perception classified by a no-report marker. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. No principle yet selects the cover, and the sheaf’s overlaps are spatial where the example’s are overlaps of content, so the proposed correlate of unity has no operational definition. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction

13 from an internal encoding). The mathematical states are spatial measures
14 and fields; identifying them with perceptual content requires independent
15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which. Phase
23 order constrains shared content only through encoders related on overlaps,
24 with a bound that grows as the square root of the population it is read
25 on. Compatible local descriptions, not synchrony, determine a unique global
26 state. A contracting self-map fixes one state but can reconstruct accurately
27 only a family of bounded diameter. Each result names a measurement that
28 phase data alone cannot supply.

29 The objection that synchrony does not fix content is older than this work.
30 Critiques of the temporal binding hypothesis argued that synchrony, where
31 it occurs, cannot say which features belong together or what they jointly
32 represent: that grouping has to be computed before synchrony could sig-
33 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
34 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
35 and a pigeonhole count. What formalization adds to the informal argument is

36 quantitative and bookkeeping content. It gives the rate at which a coherence
37 bound on content weakens with population size and the population at which
38 it stops constraining anything, the diameter within which one contracting
39 map can reconstruct, and the number of codes a separated family of states
40 requires. It also gives a complete inventory of the premises a synchrony-to-
41 content argument consumes, each of which names a measurement, together
42 with one model in which all of them hold at once.

43 The familiar noisy mean-field oscillator supplies a stationary coherence
44 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
45 condition for a declared actuator, bounds that connect phase order to con-
46 tent only through specified encoders, and conditions under which compatible
47 descriptions glue and an accurate reconstruction fixes the glued state. A
48 Lean 4 development checks these implications and a joint witness satisfies
49 their eight connecting hypotheses in a specified model. It proves neither that
50 cortex satisfies those hypotheses nor that the glued state is experienced. The
51 latter identification motivates the framework and needs its own endpoint.

52 The empirical question is which measurements could support or discon-
53 firm the proposed connections. A phase resultant can miss a locally or-
54 dered winding field; stationary phase distributions can leave the coupling
55 mechanism unidentified; and coherent adaptation can lose alignment with
56 a designated content template. Independent decoders must identify shared
57 quantities before their agreement can be tested. We therefore distinguish sta-
58 tionary branches from finite observed onset, and compare prespecified field

59 and connectivity onset models on held-out emergence recordings. The com-
60 parison tests a specified spatial prediction, not the presence of experience.
61 Section 10.2 states the comparison that bears on conscious content directly.
62 Sections 2–5 develop the three requirements; Section 6 applies their mea-
63 surement obligations to a digital candidate. Section 7 states what the formal
64 composition establishes, and Sections 8–9 examine observables and the cor-
65 tical test. Table 1 lists the connecting hypotheses the article argues from;
Supplemental Material gives the full chain, derivations and controls.

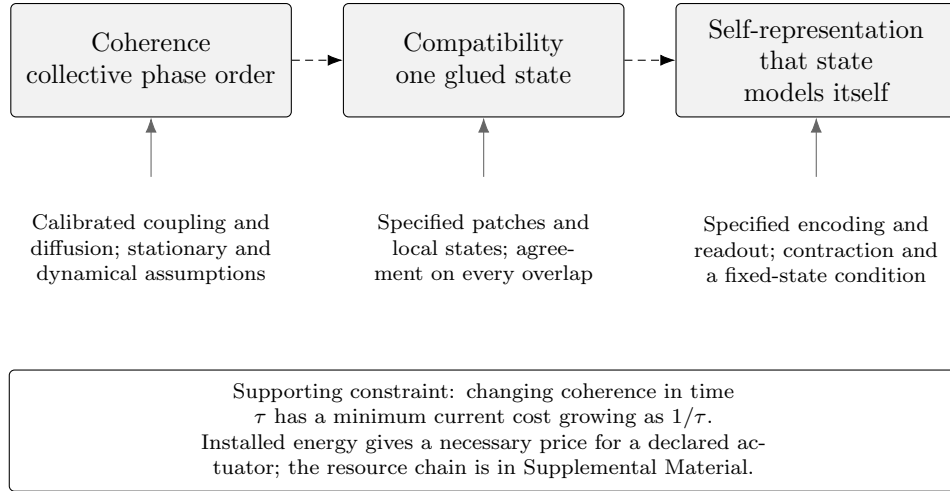


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

67 2 Limits of phase order

68 Several limits hold without the cortical hypothesis. Equal phases can coexist
69 with incompatible local descriptions; phase order constrains shared content
70 only through encoders related on overlaps. A winding field can have zero
71 global resultant while remaining locally ordered. A constant readout can
72 have a unique fixed point without reconstructing its input. A region that
73 does not distinguish relevant states cannot encode them, and a site cannot
74 report a change outside its causal past by a declared deadline. These lim-
75 its reflect familiar phase oscillator and information constraints (Acebrón et
76 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
77 The sections below state each one beside the condition or measurement it
78 motivates.

79 3 Coherence: physical coupling and its cost

80 3.1 Dynamics and the stationary threshold

81 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

82 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
83 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,

84 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
 85 quency heterogeneity and diffusion are distinct parameters. The stationary
 86 calculation below uses identical frequencies and this scalar coupling. Ap-
 87 plying it to a spatial kernel requires an aggregation rule and a check of the
 88 resulting approximation.

89 For identical frequencies in their rotating frame, the positive normalized
 90 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

91 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 92 mal derivation uses the absence of stationary current for a periodic gradient
 93 drift to obtain this density, then verifies normalization and self-consistency.
 94 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 95 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 96 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

97 Figure 2 shows this stationary branch.

98 Linearization about the uniform density gives rate $K/2 - D$ for the first
 99 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 100 exactly above threshold. About the coherent density, the quadratic form
 101 is strictly dissipative transverse to collective rotation, whose rate is zero.

102 These linear estimates concern the limiting equation. Relating them to finite
 103 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 104 tion; exchange symmetry alone does not establish the needed independence
 105 (Sznitman, 1991).

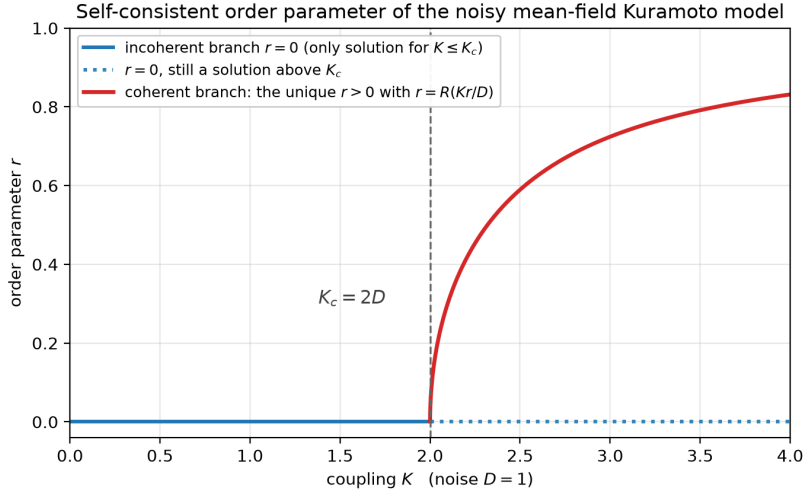


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

106 3.2 A speed limit on changes of coherence

107 One thermodynamic statement bears on coherence directly. For a smooth
 108 positive periodic phase density with probability current J and diffusion D ,

109 the current functional $\sigma_J = \int J^2/(D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

110 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
111 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
112 has no positive floor. Read the other way, a current budget C sets a minimum
113 duration $\tau \geq [\Delta \arcsin r]^2/(DC)$ for a given change. This bears on ignition-
114 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
115 2014), and on recovery latency after anaesthesia: each has a floor set by the
116 current available to drive it. The bound is a heat or power statement only
117 when J is identified with a physical flux, a reservoir at a stated temperature
118 is specified and local detailed balance holds for the steps concerned; only
119 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
120 Shenker, 2000). Installed energy supplies a separate, necessary price: for
121 a declared actuator whose response modes carry installation prices, mean
122 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
123 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$, and no cortical value of κ
124 is known. Supplemental Material gives both derivations, the register and
125 feedback budgets, and the sensitivity of the metabolic comparison to its
126 assumptions.

127 4 Compatibility: from coordinated activity 128 to a global description

129 4.1 Why common timing is insufficient

130 For overlapping patches U and V , compatibility means that restricting either
131 local description to $U \cap V$ gives the same object. Descriptions can agree on
132 shared quantities while contributing different information elsewhere. Com-
133 patible sections of the specified sheaf have a unique global extension. On
134 a finite cover, local finite spatial measures agreeing as measures on every
135 overlap also have a unique finite-measure extension whose sheaf image is the
136 glued section. No mass-one normalization is imposed.

137 In the cup example, visual and reaching descriptions must express a
138 shared quantity, such as cup location, in common coordinates. Restriction
139 retains that quantity while dropping details particular to each description.
140 Agreement permits surface appearance and hand configuration to contribute
141 to one situation. The mathematical counterexample assigns the same phase
142 to every patch while two local mass profiles disagree on a shared site. Those
143 profiles have no common extension. Because they are chosen independently
144 of phase, synchrony places no constraint on their disagreement.

145 This distinction limits the biological interpretation. Receptive territories
146 and content domains must be related to spatial patches, and the descriptions
147 must support the required extension. Matching overlap marginals of de-

148 coded probability laws need not determine a unique global joint distribution
149 (Abramsky & Brandenburger, 2011).

150 The cover is itself a choice, and the construction constrains it in one way
151 only: synchronized patches agree where they overlap. Where they do not
152 overlap there is nothing to compare, so a cover by pairwise disjoint patches
153 satisfies that obligation for *every* assignment of local descriptions and phases
154 and the gluing theorem still returns a unique global section, which is the
155 tuple it was handed. Membership of the class then reports nothing about the
156 system beyond the patches covering it. What a gluing claim asks of a cover
157 is therefore a non-degeneracy the theorem does not assume: that distinct
158 patches meet, and that what they share is a quantity fixed independently of
159 the agreement it is being used to establish.

160 The cup example exposes a second gap, between two kinds of overlap.
161 The sheaf is built over a spatial substrate, so its patches overlap where they
162 share sites. Visual and motor territories that agree on cup location overlap
163 in what they describe, not in where they lie, and the construction supplies
164 no relation between the two. One route recasts the construction over content
165 variables: each region becomes a partial decoder whose domain is the set of
166 variables it estimates, and two regions overlap exactly where they estimate
167 a common variable. The gluing argument needs only a cover and restriction
168 maps, so it applies formally to such a cover, but no result here is stated
169 for one, and which variables and decoders form it is an empirical question.
170 Until a principle selects the cover, the proposed correlate of unity has no

171 operational definition: gluing on a declared cover is proved, and which cover
 172 cortex realizes is open.

173 4.2 When coherence does constrain content

174 A declared relation between phase and content gives a positive result. Let
 175 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 176 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 177 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 178 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 179 sites; their equality on overlaps is a supplied relation between descriptions of
 180 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 181 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 182 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.
 183 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1 - r^2)}. \quad (5)$$

184 Here discrepancy is the norm of the difference, and r and N are read on
 185 whichever population the bound is applied to. That choice is what decides
 186 whether the estimate says anything. A patch carries an order parameter
 187 of its own, and reading the bound there carries that patch's resultant and
 188 that patch's site count, so a large population observed through small patches
 189 is constrained by the small count. Read instead on the whole array, the

190 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
 191 from an average over N sites. The order cannot be improved in general:
 192 one oscillator in antiphase among N leaves $N(1 - r^2)$ near 4. At $r = 1$ the
 193 observations agree exactly at either scale. A shared scalar encoder is a special
 194 case. Dropping the overlap relation permits disagreement even at perfect
 195 locking. A common nonexpansive linear projection preserves agreement, but
 196 agreement of a projection does not establish agreement of the full vector.

197 What Eq. (5) bounds is a chord, which never exceeds 2, so the estimate
 198 constrains anything only where the right-hand side falls below that. At the
 199 sheet's own patch order, the bound is informative for at most 10 sites; beyond
 200 that, the coherence-to-content link requires the bridge assumption E78 of
 201 the cortical hypothesis (Table 1 gives the full composition and its edges).
 202 Supplemental Material provides the detailed derivation of this saturation,
 203 and states three conditional derivations that replace the uniform factor $\sqrt{N}$
 204 by a count of disagreeing pairs, a chain across a cover's shared sites, or a
 205 spectral gap at a frequency-locked configuration.

206 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 207 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 208 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 209 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

210 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-

211 duce an initial mismatch. Approximate observations leave a nondecaying
 212 term in the error bound. Taking a uniform bound on Eq. (5) connects this
 213 update to the phase process.

214 4.3 What approximate agreement determines

215 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 216 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 217 Weights that vanish outside their patches and sum to one at each site select
 218 a global profile by weighted averaging. The selection lies within ε of each
 219 patch there, and any two such selections differ by at most ε , independently
 220 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 221 $L\varepsilon$ across selections.

222 At zero tolerance, the selection is the unique exact extension. At every
 223 positive tolerance there are families with no exact extension; selection then
 224 gives an approximation whose dependence on weights is bounded. Supple-
 225 mental Material gives a three-site witness attaining the bound. The mass-
 226 profile construction uses the scalar case of the content dynamics and addi-
 227 tionally requires nonnegativity of the initial profiles and observations. The
 228 general vector result supplies no identification with the spatial-measure sheaf.

229 These constructions concern declared objects: a specified cover, contents
 230 valued in a normed space, and encoders related on overlaps. They select
 231 neither the neural cover nor the decoder, and identifying their contents with
 232 perceptual content, or establishing that the identification bears on experi-

233 ence, is a further modelling task; Section 8 sets out what a measurement of
 234 it would have to establish.

235 **5 Self-representation: a state that models it-** 236 **self**

237 **5.1 The reflexive construction**

238 Compatibility gives a global state. Reflexivity relates it to an encoding within
 239 that state. Equip global sections with a complete metric. If the additional
 240 restriction-resonance condition identifies encoding with restriction to a region
 241 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

242 The region and readout are fixed across the state space. If F contracts and
 243 fixes the section s_* formed from the local descriptions, Banach's theorem
 244 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 245 (Table 1 gives the full composition and its edges) supplies the contraction
 246 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
 247 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
 248 a relaxation time $\tau > 0$. The factor is stipulated, not derived. It borrows
 249 the rate $(K - 2D)/2$ at which the first harmonic of the uniform density

grows (Section 3), a rate about a different state from the one F acts on,
and no result derives a Lipschitz law for an encoding or readout from the
phase dynamics. Constructing a map with that factor remains a modelling
obligation. A fixed contraction can reconstruct accurately only a restricted
family of states: if its error is at most ε throughout that family, any two
members are at most $2\varepsilon/(1 - \Lambda)$ apart.

In the cup example, the global state includes the represented scene and the
perceiver’s relation to it. A local encoding must retain enough information
for the readout to recover that modelled state; cup position alone would be
insufficient if hand configuration and viewpoint could not be reconstructed.

For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruction gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
fixed positive ε and τ . Exact uniqueness survives while the error guarantee
weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an
encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that

268 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

269 Thus M relevant states pairwise more than 2ε apart require at least M
 270 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 271 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 272 existing self-prediction map.

273 This is an additional accuracy criterion on a family, with no requirement
 274 for multiple exact fixed points. For the same fixed contracting map used
 275 by E89, however, the diameter bound above is an additional restriction: at
 276 zero error the family contains at most one metric-distinct state. A broad
 277 family therefore requires a different, input-conditioned reconstruction map,
 278 one map F_u with its own fixed point for each input u , with its own accuracy
 279 and stability assumptions. No such map is formalized or constructed here.
 280 Under E89 the proposed minimal-self correlate is therefore a condition on
 281 one self-state, not a model of a self that follows a changing scene. A blind
 282 encoder or constant readout fails the stated criterion for separated states
 283 despite having a unique fixed point.

284 Where the encoding is read as the state's own restriction to a region, the
 285 criterion becomes a condition on the region. Restriction resonance requires
 286 the encoder and that restriction to identify the same pairs of states, so a
 287 boundary reconstructing two relevant states more than 2ε apart while its

288 region reads them identically is not resonant. Read forwards, a resonant
 289 boundary that reconstructs a pairwise separated family restricts injectively
 290 on it. Which states a region reads identically is fixed by where the region
 291 sits, before an encoding is chosen, so the resolution the family demands is
 292 the region's and no readout supplies it. The implication runs one way: a
 293 region that does resolve the family is not thereby shown to carry a resonant
 294 encoding, and E89 supplies no such resolution.

295 Resonance and resolution are jointly satisfiable without a local fold. Take
 296 the region to be the whole substrate and use the identity readout: restric-
 297 tion is the identity, so this boundary is resonant and reconstructs every de-
 298 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 299 E89's strict contraction premise on a nontrivial metric space. A properness
 300 condition on the region is therefore a separate requirement, and the fixed-
 301 contraction diameter bound remains in force when E89 is also required. With
 302 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 303 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 304 so the family is no larger than the δ -packing number of what the region car-
 305 ries. That is the shape a continuous family of states needs, and at the whole
 306 substrate it says nothing, because the region's range is everything. The gain
 307 L is declared and derived here for no physical readout, and at unbounded
 308 gain the bound says nothing.

309 **5.3 Agreement after a change takes communication**

310 Snapshot compatibility does not specify how quickly descriptions respond
311 when a shared quantity changes. In a synchronous network whose updates
312 read only the site’s own state and declared neighbours’ messages, two initial
313 configurations agreeing on a site’s causal past through T rounds produce the
314 same state there at round T . The deadline consequence is the one Section 2
315 states, and it follows from the reconstruction bound in Section 5.2: a report
316 from that site is a readout of a code that the two interventions leave identical.

317 A three-site delay line witnesses failure after one round and exact report-
318 ing after two; cutting the path prevents delivery at every round. A report
319 may still happen to be correct about one fixed world. The bound concerns
320 responsiveness across interventions.

321 Throughout this section *prediction* names a reconstruction map, and the
322 relevant family of states, the metric, the tolerance, the communication graph
323 and the deadline are all declared inputs, as are the represented variables and
324 the fidelity counted as accurate. Reading the map as temporal neural pre-
325 diction would require dynamics; reading the resulting code count as physical
326 memory capacity would require a hardware model, which a state-dependent
327 readout would have to extend.

328 6 What the conditions ask of a digital candi- 329 date

330 A claim that a digital system, such as a large language model (LLM) run-
331 ning on graphics processing units (GPUs), has unified content or represents
332 itself commits its maker to the measurements this account asks of cortex.
333 The maker must identify the shared quantities the system’s parts describe,
334 show that decoders fitted independently on those parts agree about them,
335 and show that a declared local code separates the relevant family of states
336 and survives intervention. The indicator-properties approach assesses arti-
337 ficial systems against computational properties drawn from theories of con-
338 sciousness (Butlin et al., 2023); the conditions here are fewer and each is
339 a measurement rather than an architectural property. A digital system is
340 also where they can be measured today, with full access to its state and
341 exact interventions on it: fitting decoders for one shared quantity on two
342 separate layers, heads or token positions, and testing their agreement and
343 whether reconstruction survives intervention, would carry out the proposed
344 measurement directly.

345 One constraint follows from the architecture alone. Within one forward
346 pass of a causally masked model, position t reads only its causal past, so an
347 intervention outside that past cannot be reconstructed at t by that step. This
348 is the deadline result of Section 5 applied to the mask, and it bounds which
349 variables a position can encode about the rest of the sequence within one

350 step. A second constraint needs a premise the architecture does not supply:
 351 a linear encoder and readout factoring through d coordinates pay reconstruction
 352 error at least the discarded spectral tail of the coupling operator, but
 353 query/key dimension does not supply such a bottleneck for softmax attention.
 354 These results rank no architecture family, imply no general inferiority of
 355 GPU hardware and settle no question of machine experience. Supplemental
 356 Material gives the bottleneck derivation, a synthetic two-view task with its
 357 controls, the information ceiling of token outputs and the hardware scope.

358 **7 The formal composition**

359 The Lean development composes the three requirements into one conditional
 360 theorem. Its passive and feedback branches each accept eight connecting hy-
 361 potheses, running from finite capacity through a register, dissipation and a
 362 continuum coupling to the field, coherence, the cover and the self-prediction
 363 fixed point. Each edge is supplied rather than derived from the one above it,
 364 and the inferential content of the composition is its last step, where contrac-
 365 tion and a fixed glued section give uniqueness. Table 1 lists the four edges the
 366 article argues from; Supplemental Material gives the full chain, its thermo-
 367 dynamic edges and the register ledger of the feedback branch. Both branches
 368 also take a cover carrying exact overlap compatibility and a reflexive bound-
 369 ary specifying encoding and readout, and conclude that the glued section is
 370 the unique fixed point of self-prediction. Composing the edges establishes

371 that the premises are consistent, named in one place and machine-checked
372 together; it does not establish that self-representation follows from coher-
373 ence. Joint witnesses satisfy every edge of each branch in a specified model,
374 and establish no common cortical mechanism. The development declares no
375 additional axioms beyond its libraries’ foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

376 8 What the observations establish

377 The composition holds under its premises. Whether observation can estab-
378 lish those premises is a separate question, and the three requirements it
379 connects concern different objects: phase dynamics, local descriptions and
380 internal encodings. Their observables must retain these distinctions. The

381 following controls show which mechanisms remain indistinguishable, which
 382 temporal approximations fail, which states one observable cannot separate,
 383 and what independent content measurements must establish. Each is a finite
 384 numerical regime on a declared model, with full parameters and controls in
 385 Supplemental Material, and none of them calibrates a cortical quantity.

386 **8.1 Stationary distributions leave the mechanism un-** 387 **determined**

388 Equation (2) gives a concentration–coherence relation with no fitted shape
 389 parameter. Estimate r from circular resultant length and a from density
 390 shape: inverting R from the measured r would make agreement automatic.
 391 Even independently estimated agreement identifies a statistical family shared
 392 by different mechanisms.

393 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 394 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 395 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 396 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 397 and diffusion D_0 have that same density. Their stationary distributions co-
 398 incide window by window despite different causes.

399 Resolved dynamics supply additional information. In the ideal model
 400 with continuous latent paths, quadratic variation accumulates at rate $2D$
 401 and conditional increment means constrain drift; a filtered observation of

the phase needs its own observation model before either is read off. Separating endogenous coupling from a matched common drive additionally requires a measured drive or controlled perturbation; identifying electromagnetic coupling requires evidence specific to that interaction. Supplemental Material gives the equivalence construction and its numerical check.

8.2 Finite ramps change the observable onset

The square-root law in Eq. (3) describes stationary solutions, and a changing coupling needs time for the phase distribution to approach them. In a deterministic mean-field control crossing the threshold, none of the scanned rates meets the declared tracking tolerance at onset, so a slow control variable supplies no general guarantee of branch tracking. In a finite-population ramp study ($N = 2000$, $D = 1$, 32 replicas, 8 speeds), faster ramps delay escape. The fitted delay exponent 0.444 has a replica-bootstrap interval $[0.398, 0.496]$ that excludes the ideal $1/2$, a deterministic reference shows that the escape criterion changes the fitted exponent, and no common asymptotic exponent is established. Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002) makes the apparent exponent depend on ramp speed and fitting window. Supplemental Material gives the sweep and its figure, the fluctuating-coupling and population-size controls, and the signed-network crossing bracket.

The ideal delayed-onset calculation motivates a square-root scaling hypothesis; its exponent is generic to suitable bifurcations and does not iden-

424 tify the coupling medium. A within-subject emergence design can vary the
 425 propofol taper and test how an operational onset changes with that manipu-
 426 lation, but interpreting the result as a coupling-rate law requires a measured
 427 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 428 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 429 describes how D could be estimated from resolved phase traces.

430 **8.3 A winding state leaves the order parameter am-** 431 **biguous**

432 The global order parameter r discards spatial arrangement. On a periodic
 433 sheet, a phase field can wind once around the sheet and have $r = 0$ while
 434 neighbouring sites remain locked. At identical frequencies, every such wind-
 435 ing is stationary for an isotropic symmetric kernel: contributions at opposite
 436 separations cancel. It violates the quarter-turn initial-spread premise of the
 437 convergence result, which is why that result can still select the uniform state.

438 In the spatial control, the winding persists below the interpolated de-
 439 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform
 440 and winding starts under the same coupling, diffusion and kernel end at
 441 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 442 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 443 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 444 realizations. These are finite controls, not a cortical wave identification or a

445 derived transition.

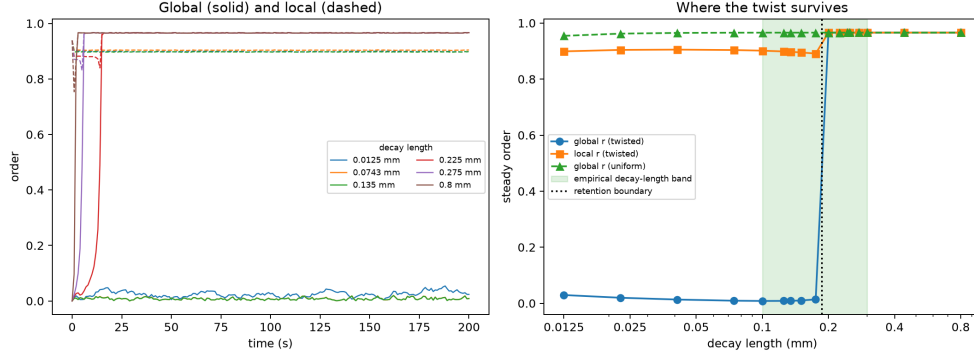


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

446 Linearization about a winding reweights each separation’s coupling by
 447 the cosine of its phase advance. The resulting symmetric kernel has explicit
 448 mode rates: nonnegative reweighted coupling gives no growing mode, while
 449 a positive rate destabilizes that winding. The criterion concerns linear rates,
 450 not asymptotic stability, and the measured retention boundary need not
 451 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 452 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 453 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 454 winding from that model even though the phase-only equations retain it as

455 a stationary state.

456 Neither model turns these states into measured cortical spirals. The pe-
457 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
458 no propagation speed, and the amplitude variable has no calibrated relation
459 to a recorded envelope. Supplemental Material gives the spectral, amplitude
460 and defect results. Locally, Eq. (5) can still bound content disagreement on a
461 winding patch when shared encoders are supplied, although its global $r = 0$
462 form is uninformative. A nonzero ring winding number q also obstructs a
463 global real-valued phase lift; it has no identified content meaning here. Thus
464 an onset criterion based only on r can register a change of basin as well
465 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
466 observable.

467 **8.4 Coherent adaptation can lose environmental align-** 468 **ment**

469 An adaptive coupling rule might supply representational structure, but its
470 objective must reward the relevant agreement. For symmetric coupling, pair-
471 wise sine terms cancel in the population mean drift. Minimizing squared drift
472 at that fixed mean therefore rewards equalizing drift across nodes. In the
473 frequency-cluster control, this favors coupling between unlike-frequency clus-
474 ters, whereas the diagnostic environmental template rewards coupling within
475 clusters.

476 The formalization proves descent of the squared-drift objective for a spec-
477 ified gradient flow at fixed phases. Its functional differs from thermodynamic
478 entropy production: at a coherent stationary density the probability current
479 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
480 rial gives the operator, its gradient and the conditions of the descent result.

481 The finite control co-evolves noisy phases and symmetric, nonnegative
482 couplings at fixed total resource. Its template is a diagnostic covariance
483 of latent signals, never a training input. Across three illustrative seeds,
484 second-half order remains between 0.9167 and 0.9222, while the within-
485 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
486 0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel con-
487 trols are reported in Supplemental Material.

488 The immediate post-update objective reduction disappears when aver-
489 aged over complete update intervals. A prespecified grid of 6 rate–cadence
490 combinations contains no case meeting the 5% sustained-reduction criterion
491 while retaining coherence and losing alignment. The finding is coherent adap-
492 tation away from the specified structure, with no demonstrated sustained cost
493 reduction.

494 Kernel alignment and overlap compatibility are different observables.
495 This control addresses acquisition of its designated template under its par-
496 ticular objective. Learning models can explicitly include data in an objective
497 (Corberi et al., 2026); an account of agreement about the cup must specify
498 the content its learning rule is meant to preserve.

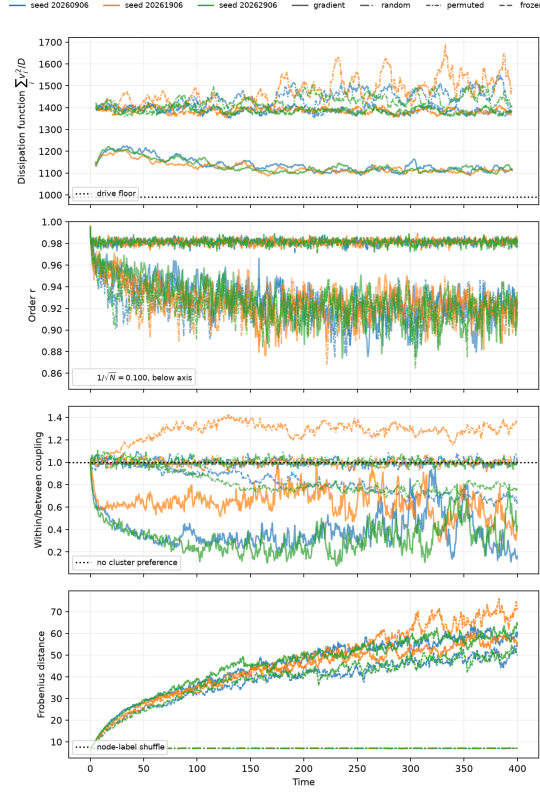


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

499 8.5 Content agreement needs independent decoders

500 A neural test needs independently identified domains, local descriptions and
501 restriction maps. Two decoders should estimate a shared content on over-
502 lapping domains before their agreement is examined. Discrepancy requires a
503 null under decoding noise and controls using deliberately compatible and in-
504 compatible examples. An implementation on synthetic data separates these
505 cases at $\text{AUC} = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
506 the compatible construction scores 0.476 on an interval containing one half.
507 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
508 bias can manufacture disagreement. Per-patch decoding diagnostics detect
509 these failures; the compatibility statistic alone does not.

510 Phase agreement cannot replace that measurement. On constructed con-
511 tent, phase carries a share 1.000 of the content entropy when content is built
512 from it and 0.000 when it is not. A phase-derived statistic reports com-
513 patibility on phase-locked patches whose local descriptions disagree. The
514 implemented content statistic detects their incompatibility. These controls
515 validate a measurement on known constructions; a neural content model and
516 decoder remain to be established.

517 Choosing the cover also chooses a candidate subject: trivial descriptions
518 can agree while carrying little content, and a partition can conceal disagree-
519 ment. The useful choice must therefore be constrained by what neural ac-
520 tivity encodes. The exact composition assumes compatibility of its supplied
521 cover. The content mechanisms above identify additional hypotheses under

522 which agreement can arise; they leave the empirical choice of content and
523 the connection to that cover explicit.

524 **8.6 Estimator calibration on exploratory EEG**

525 The concentration range and the dependence structure decide whether the
526 concentration-coherence relation can be tested at all. Under the specified
527 log-density estimator and bias-floor criterion, discrimination at $a = 0.542$
528 requires 3000 independent sites in the scanned designs, rising to 31000 when
529 sites share a mean in groups of 10. These counts describe that estimator and
530 dependence model at a declared design concentration; the intended recording
531 requires its own calibration. Supplemental Material reports the bin-count
532 controls and the full dependence-aware design.

533 The estimators were also exercised on public propofol-sedation EEG (Ba-
534 jwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels,
535 band-passed, Hilbert-transformed and pooled over pairs and 100-ms win-
536 dows. Across 8 subjects the pooled concentration spans 0.087–0.142, and
537 the residual from the Bessel relation is small. That agreement is uninforma-
538 tive. Pooling over a window does not remove the common rotation of the
539 phases, so a perfectly phase-locked 10-Hz signal passed through the same
540 pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its
541 instantaneous spatial resultant is one. The bipolar montage also removes
542 spatial coherence: a field whose raw channels have instantaneous resultant
543 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform

544 pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel
545 ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks
546 the estimator’s behaviour and establishes no spatial concentration, coher-
547 ence level or test of the stationary density. Sedation recordings also leave the
548 awakening hypothesis untested.

549 An informative dataset therefore needs an estimand that removes the
550 common rotation, a montage whose effect on spatial phase is modelled, suf-
551 ficient concentration range or precision, calibrated estimators and an ob-
552 servation model for dependence and source mixing. Intracranial recordings
553 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
554 informative concentration range, but that range must be measured. The
555 design counts above are calculations for the specified estimator, not sam-
556 ple requirements transferable to another recording. Public mouse calcium
557 imaging provides thousands of spatially located neurons across waking, sleep
558 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
559 phase-estimation band used here. Site count, state coverage and bandwidth
560 are separate requirements.

561 **9 The cortical hypothesis and an awakening**

562 **test**

563 The cortical proposal identifies a physical source of coupling and asks whether
564 its changes help explain recovery. The tests in Section 8 determine what

565 observations must distinguish before a phase trace can support this identifi-
566 cation.

567 **9.1 Physical identification and calibration**

568 The cortical hypothesis identifies part of effective coupling with endogenous
569 extracellular electromagnetic interactions. Fields can modulate spike timing
570 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
571 propose coordination of distributed neural information (Pinotsis & Miller,
572 2022; McFadden, 2020). Phase organization under propofol motivates study-
573 ing this regime (Bastos et al., 2021), but does not measure coupling.

574 The formal field predicate requires a jointly continuous kernel, a normal-
575 ized substrate, coupling equal to the kernel average, and no single site pro-
576 viding more than half that strength. Toy constructions satisfy it. A cortical
577 realization must specify the scale at which the kernel and its reduction ap-
578 ply: calcium imaging reports different state-dependent modularity patterns
579 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

580 Cortical columns are the obvious candidate unit, and they answer one
581 of the two demands the formalism makes. As the oscillator index they are
582 a serious proposal: they are delineated by response properties and anatomy
583 rather than by the agreement they would be used to establish (Mountcastle,
584 1997), and a phase can be estimated from a column-sized territory at the
585 electrode spacings used for cortical wave measurements (Townsend et al.,
586 2015). As the content cover they are a poor one, for three separate reasons.

587 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
588 the compatibility obligation without constraining anything (Section 4). The
589 cover condition additionally requires the patches to exhaust the substrate,
590 while much of association cortex has no accepted columnar parcellation. And
591 the column's standing as a canonical cortical unit is contested (Horton &
592 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
593 supplies the oscillator index; the content cover remains an open empirical
594 question, which the bridge assumption E78 records.

595 The spatial reduction is also testable. A row-sum aggregation at zero
596 diagonal tracks the uniform-coupling control over the tested spatial decay
597 range, while frequency heterogeneity displaces the operational threshold. The
598 fit window biases both estimates relative to $2D$; agreement with the control
599 is therefore the supported conclusion. Supplemental Material reports the
600 reduction, its numerical error and the distinct assumptions needed to identify
601 the kernel average with the scalar coupling of the installed-energy condition.

602 The field commitment requires physical evidence beyond a choice of math-
603 ematical representation, and the available support comparisons do not supply
604 it. A finite matrix spread over positive-mass cells preserves its total cou-
605 pling and phase correlation exactly, while point-supported vanishing under
606 an atomless measure and a disadvantage from excluding a better-correlated
607 pair concern specified support constraints; Supplemental Material reads the
608 same comparisons against a digital candidate. Neither supplies energy prices.

609 A spike-time shift alone supplies no coupling rate. The dimensionless

product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

This rate describes how induced timing shifts accumulate into drift. Neither it nor the installation factor κ follows from a heat bound. Testing the physical threshold requires independent estimates of the interaction kernel, its aggregation and diffusion in the same cortical regime.

9.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (10) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

629 For a smooth crossing at nonzero speed, evaluating the stationary branch
630 along the changing control gives a deterministic tracking approximation that
631 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

632 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
633 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
634 proximation under the tested initialization. A separation of geometry and
635 phase time scales can support tracking away from threshold. The awakening
636 hypothesis needs a calibrated dynamical prediction for an operational onset
637 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
638 recovery latency.

639 **9.3 A protocol for the awakening experiment**

640 Neural inertia describes a behavioural barrier and hysteresis between anaes-
641 thetic induction and emergence (Friedman et al., 2010); recovery can pass
642 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
643 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
644 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
645 mechanism from a synaptic one. A possible discriminator is spatial onset
646 relative to independently estimated local field coupling and network connec-
647 tivity. The present mean-field model has no spatially varying κ and derives

no nucleation order, so a field-based onset map needs a separate model specified before the observations are inspected.

For that comparison, partition the recording into prespecified regions i and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements, and a connectivity score c_i from a separately estimated structural network. For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

Predicted onset is the first crossing of a region-specific threshold fixed from pre-onset fluctuations. Fit the two scales on training sessions, then compare held-out onset times and site order under the same observation model. Larger field scores predict earlier onset only when baseline order and diffusion do not reverse the ranking. This declared approximation omits interregional coupling and saturation; a successful ranking would support its field covariate, not prove ephaptic causation. Supplemental Material specifies the controls and failure conditions.

A within-subject protocol can manipulate emergence rate, with spatially resolved recordings to compare the proposed field and connectivity models:

1. **Manipulate emergence rate.** Vary the propofol infusion taper

668 across sessions within the same subject, recording drug concentrations
669 or a validated pharmacokinetic proxy to quantify the manipulation.

670 2. **Measure behavioural recovery.** Record time-to-response to a stan-
671 dard stimulus as the independent behavioural endpoint, rather than
672 estimating ΔK directly.

673 3. **Resolve spatial onset.** Identify anatomical locations where coher-
674 ence first crosses the baseline-scaled threshold and score the two maps
675 of Eq. (12) on held-out sessions. The two covariate maps are fixed
676 before the outcome sessions are inspected.

677 4. **Match physical controls.** Confirm that the observed coherence-to-
678 amplitude relation holds under the specific local diffusion D and ob-
679 servation noise. The model’s quadratic-variation rate $2D$ applies to
680 the latent phase; a band-limited analytic phase is smooth below its
681 correlation time, and on a constructed field its squared-increment rate
682 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
683 fore requires an explicit observation model for filtering and sampling.

684 Predeclare the recording window, regional fluctuation thresholds, and a
685 shared timing-error model. Fit the two scales and that observation model
686 on training sessions only; on held-out sessions, score the observed site-onset
687 times and sites that never cross by the same regional predictive rule. Report
688 site order as a secondary outcome and behavioural time-to-response sepa-
689 rately. Sessions without a crossing are censored observations, not zero delays

690 or exclusions. Compare the two scores only where independently estimated
691 field and connectivity maps predict distinguishable onset patterns at the
692 recording resolution. A held-out field-model advantage would support that
693 specified spatial predictor; it would not identify an ephaptic cause, shared
694 content or experience. Supplemental Material gives the controls and pilot
695 quantities needed to judge whether the comparison is informative.

696 **10 Discussion**

697 **10.1 Unity and a minimal self as interpretive motiva-** 698 **tion**

699 We propose the state glued from compatible local descriptions as a formal
700 correlate of conscious unity, and its accurate internal reconstruction as a cor-
701 relate of a minimal self. In the cup example, this would mean one description
702 that preserves the cup's relation to the perceiver as well as the scene. Com-
703 patibility permits distinct contents and uncertainty; the shared quantities
704 must be chosen from independent evidence about what each region encodes.
705 The reconstruction need not be a verbal or autobiographical account.

706 The minimal self, in Gallagher's sense, is the pre-reflective sense of be-
707 ing the subject of an experience, present before any narrative or reflective
708 self-description (Gallagher, 2000). Two of its features motivate the proposed
709 correlate. It is not a separate object of experience but a perspective built

710 into the experienced scene; correspondingly, the reconstruction is not a sec-
 711 ond representation beside the glued state but a map whose fixed point is that
 712 state, read through a part of it. It is immediate rather than inferred from
 713 a higher-order state; correspondingly, the fixed point needs no further state
 714 that represents the first. The correspondence is structural and does not show
 715 that a fixed point is experienced as a perspective. Self-model theory locates
 716 the phenomenal self in a transparent model a system has of itself (Met-
 717 zinger, 2003), and interoceptive-inference accounts ground it in prediction of
 718 the body’s own states (Seth, 2013). The present condition is narrower than
 719 either: it asks only that a declared local encoding recover the global state,
 720 including perceiver-relative variables, at a declared fidelity, and it says noth-
 721 ing about transparency or about which bodily variables must be modelled.
 722 Either account could supply the family of states and the fidelity that the con-
 723 dition leaves as inputs. Under E89 that family is one state at zero error and
 724 a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Section 5.2), so the
 725 condition as proved constrains a single self-state; a self that persists through
 726 change needs the input-conditioned map that is not constructed here. Its
 727 proposed relation to experienced perspective remains an interpretation. Nei-
 728 ther identification is among the connecting hypotheses of the composition
 729 (Table 1), and satisfying them would not measure experience. The unity
 730 proposal also lacks an operational definition. The gluing theorem holds for
 731 whatever cover it is given, a disjoint cover satisfies it trivially, and the cover of
 732 interest overlaps in content rather than in space (Section 4); until a principle

fixes that cover, the correlate is a schema awaiting one.

10.2 A comparison that could disconfirm the link to content

The framework predicts that conscious content tracks compatibility, not coherence. The comparison must also separate compatibility from how well each region encodes the stimulus, because two accurate decoders agree automatically. If decoders d_A and d_B estimate a shared stimulus quantity x with errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

Consciously perceived stimuli are usually decoded better, so higher agreement on those trials would follow from single-region fidelity alone. At matched single-region error, agreement differs only through the covariance of the two errors, which is the term compatibility bears on: two regions describing one content should err together.

Present matched stimuli that are perceived on some trials and not on others, for example at masking or rivalry threshold, and fit decoders for one shared stimulus quantity independently on two regions that both encode it. Classify trials by perceptual state with a no-report measure, such as reflexive eye movements or pupil responses that track the dominant percept in rivalry (Tsuchiya et al., 2015), so that the classification does not also

752 select for the neural correlates of reporting; report-based classification is a
753 secondary analysis. Match the two classes on phase coherence between the
754 regions and on each region's own decoding error (or, where the stimulus
755 value is uncertain, on each region's signal strength). The statistic is the
756 excess agreement over a fidelity-matched null: pair region *A*'s decode on one
757 trial with region *B*'s decode on another trial of the same stimulus and error
758 stratum, which preserves both marginal accuracies and removes any coupling
759 of the errors.

760 The proposed link predicts a larger excess on perceived trials at matched
761 coherence and matched single-region accuracy (Fig. 5). Equal excess across
762 classes, or excess that follows coherence and not perceptual state, counts
763 against it. A positive result is not yet specific: noise shared upstream of both
764 regions also correlates their errors, so the excess should be compared against
765 an upstream region's decoding error, matched in the same way. A self-model
766 comparison has the same form: at matched accuracy for the scene variable,
767 the reconstruction of perceiver-relative variables, such as hand position rel-
768 ative to the target, from a declared local encoding should be more accurate
769 on perceived trials. These are correlational tests of proposed correlates, and
770 the no-report classification is itself an indirect marker of experience.

771 **10.3 What a test of the full proposal must establish**

772 A test needs separate estimates of effective coupling and diffusion, shared
773 content on overlapping regions, and reconstruction from a specified encoding

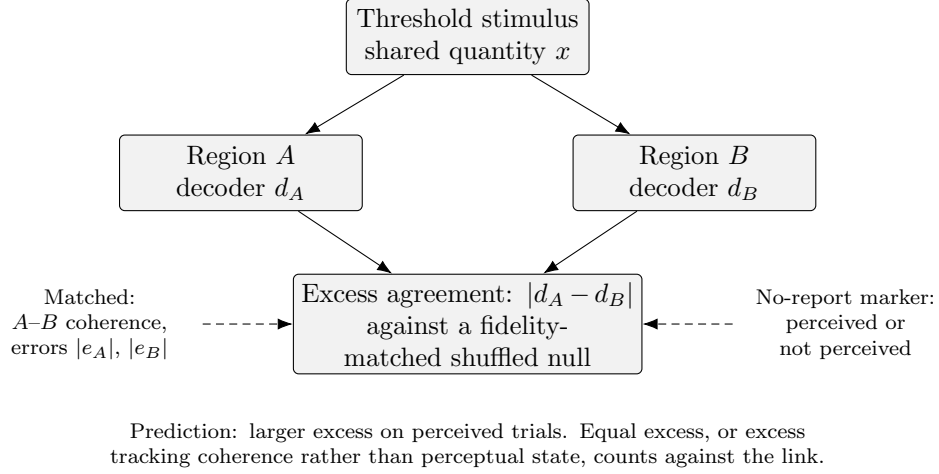


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

774 at a declared fidelity. The cover, relevant state family and decoder must
775 be fixed without selecting them for agreement. An independent experiential
776 endpoint is needed to evaluate the interpretation; absence of a report alone
777 does not define absence of experience. A unified episode with incompati-
778 ble decoded descriptions would challenge the unity identification. A unified
779 episode lacking the proposed self-encoding would challenge the minimal-self
780 claim. Conversely, satisfying the mathematical conditions without such an
781 episode would challenge their sufficiency as correlates.

782 For the physical route, cortical actuator modes and prices, the geometry–
783 coupling relation and a dynamical onset model remain to be measured. The

784 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
785 diction. Controlled changes in coordination and independently decoded con-
786 tent agreement would address the representational connection. The synthetic
787 decoder exercise calibrates part of these measurements, and the EEG exercise
788 identifies what a pooled bipolar estimand cannot measure; neither identifies
789 a neural content code or tests experience.

790 **10.4 Explanatory scope**

791 Integrated-information, global-workspace and predictive-processing accounts
792 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
793 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
794 relations among coupling, local content and reconstruction. Its formal field
795 condition requires a continuous kernel, a normalized substrate, a specified
796 average coupling and noise, and a site-contribution bound. Those conditions
797 do not identify an electromagnetic substrate; the extracellular interpretation
798 needs the separate calibration of Section 9. The joint witness shows that
799 the formal hypotheses can hold together in a declared model. A cortical
800 realization and an experiential interpretation require additional evidence.

801 Four limitations bound every claim above. Synchrony is not itself an
802 experiential measure, and nothing here establishes a consciousness criterion:
803 the conditions are proposed correlates to be tested against report or another
804 independent endpoint. The present EEG analysis checks a measurement
805 pipeline and does not validate any experiential claim. The formal results are

806 conditional on declared encoders, covers and maps whose cortical counter-
807 parts are unmeasured. No principle selects the cover on which compatibility
808 is assessed, so the unity correlate is not yet operationally defined.

809 The remaining explanatory question is whether these relations occur in
810 neural activity and matter for experience. The framework identifies candidate
811 correlates and tests of their physical and representational premises. It does
812 not explain why satisfying them would be accompanied by experience.

813 11 Conclusions

814 Phase coherence, compatibility and self-reconstruction are separable condi-
815 tions. Coherence constrains shared content only through encoders shared on
816 overlaps and with a bound that grows as the square root of the population
817 read. That bound is vacuous beyond 10 sites at the simulated sheet’s patch
818 order, so for populations of cortical size the positive link from coherence
819 to content is the bridge assumption E78 and not a derived result. Com-
820 patible local descriptions, not synchrony, determine one global state; and a
821 contracting self-map fixes that state while reconstructing only a family of
822 bounded diameter. Each result replaces an assumption about synchrony by
823 a measurement: independently decoded agreement on overlaps, a declared
824 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
825 ison of Section 10.2 states how such measurements could count against the
826 proposed link to conscious content.

Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material's convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from

Table 2 (continued)

Symbol	Defined	Meaning
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

828 Data and code availability

829 The Lean development, simulation scripts, tests and saved numerical sum-
830 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
831 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-
832 lib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication
833 simulation values are generated from saved summaries without rerunning

834 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
835 ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
836 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
837 and selection are described in Supplemental Material and implemented in
838 `simulations/empirical_collapse.py`. The repository contains the repro-
839 ducible analysis and does not redistribute the raw recordings.

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843 **Use of AI tools**

844 AI assistance to this work was substantial and is not separable by task.
845 Several coding assistants and chat interfaces were used in an interleaved way
846 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
847 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
848 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
849 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
850 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
851 they contributed to the Lean development, the simulation and analysis code,
852 and the drafting and revision of both documents. Because the tools were

853 used in combination and in alternation, no section, proof, figure or result is
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855 responsible for all content, including everything a tool produced. Machine
856 checking establishes the stated Lean results under their hypotheses; it does
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